

Project-Based Learning (PBL) Report

Comparative Technical Analysis of a WordPress-Based Dynamic Cricket Website vs Sportskeeda, Cricbuzz, and ESPNcricinfo (March 2026)

Title:

Comparative Technical Analysis of a Dynamic Sports (Cricket) Website Built Using WordPress CMS vs Industry Platforms (Sportskeeda, Cricbuzz, ESPNcricinfo)

Course / Subject: Content Management System (Project-Based Learning)

Student :	1. Shivraj Mule	ROLL NO 34	PRN 2514110456
	2. Vaibhav Anand	ROLL NO 121	PRN 2514111414
	3. Ayush Kumar Singh	ROLL NO 35	PRN 2514110457
	4. Amit Gadade	ROLL NO 21	PRN 2514110443

Project Website URL: sportly-transiverse.xo.je

Department: INFORMATION TECHNOLOGY

Class / Division B. TECH IT- SEM II DIV-1

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Abstract

This report presents a comparative technical and architectural evaluation of a project-based learning (PBL) dynamic sports website developed using a custom WordPress theme and hosted on InfinityFree, benchmarked against three large-scale sports platforms: Sportskeeda, Cricbuzz, and ESPNcricinfo. The analysis compares the systems across architecture & tech stack, UI/UX and content strategy, and performance metrics derived from PageSpeed/Lighthouse audits (industry averages for March 2026).

The PBL website demonstrates a complete CMS-driven workflow: theme conversion from static HTML to WordPress templates, dynamic content rendering using the WordPress Loop, REST API integration for live cricket data, plugin-based performance and security hardening, and resolution of WordPress routing issues such as 404 errors and permalink misconfiguration. In contrast, the competitor platforms employ React/Next.js SSR, CDN edge caching, reverse proxies, distributed backend services, and high-concurrency infrastructure to support real-time updates, SEO at scale, and monetization via ad-tech ecosystems.

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1. Introduction

Modern sports websites typically fall into two engineering categories:

CMS-driven publishing platforms (optimized for content creation workflows, editorial updates, and templated rendering), and

real-time sports data platforms (optimized for live scoring pipelines, frequent updates, and high traffic concurrency).

The PBL website belongs primarily to the first category, enhanced with live cricket data integration using REST APIs. Meanwhile, Sportskeeda, Cricbuzz, and ESPNcricinfo represent production-grade platforms designed for high traffic, real-time updates, SEO visibility, and monetization.

Objective of this report:

To evaluate the strengths, limitations, and practical engineering differences between a WordPress-based dynamic sports website and enterprise-scale sports platforms.

2. Project Overview: PBL WordPress Website

2.1 Project Title

Development of a Dynamic Sports Blog Website using WordPress

2.2 Objective

To design and develop a sports website that: - displays sports content for Cricket, Football, and Basketball, - allows dynamic content management via the WordPress dashboard, - supports live match data integration through external APIs.

2.3 Technologies Used

WordPress (CMS)

PHP (theme templates and server-side rendering)

HTML, CSS, JavaScript (custom UI and interactivity)

REST API (live sports data integration)

InfinityFree Hosting (deployment)

2.4 System Architecture (PBL Website)

Conceptual flow:

User (Browser)

- Frontend UI (Custom HTML/CSS/JS)
- WordPress Backend (PHP + Theme Templates + WP Core)
- Caching Layer (transients / REST cache concept + optimization plugins)
- Data Sources (WP Database + External Sports API)

This architecture is appropriate for a CMS-based portal because it supports content updates, templating, navigation management, and external API-based dynamic widgets.

2.5 Problems Faced & Solutions Implemented

Problems Encountered

1. Pages showing 404 errors
2. Hardcoded links failing after WordPress integration
3. Static design not fetching dynamic dashboard content
4. Navigation menu not functioning properly
5. WordPress routing issues

Solutions Implemented

- Updated permalink structure to "Post Name" to fix rewrite/routing issues
- Converted static HTML into WordPress theme templates: - front-page.php, page.php, single.php, header.php, footer.php - Implemented WordPress Loop for dynamic post/page rendering
- Rebuilt navigation using WordPress menu system
- Removed hardcoded .html links and used WordPress functions such as: - home_url()
- get_template_directory_uri()
- Integrated external sports API for live match data
- Resolved 404 issues using permalink configuration and .htaccess rewrite rules

2.6 Plugins Used (Functional Subsystems)

Performance & Caching - LiteSpeed Cache

- WP REST Cache
- Smush
- WP-Optimize

Security - Wordfence Security

- Akismet Anti-spam

SEO + Analytics - Rank Math SEO
- Site Kit by Google

Operations (Backup/Staging/Migration) - All-in-One WP Migration and Backup
- WP STAGING and backup-related tools

UI Tooling - Elementor (used for layout building; can increase asset weight if overused)

3. Methodology

3.1 Data Collection Approach

This report uses: - Technology stack signatures obtained via Wappalyzer/BuiltWith (March 2026 snapshot provided).
- Performance audit metrics based on PageSpeed Insights / Lighthouse average estimates for March 2026.

3.2 Evaluation Criteria

The platforms are compared using:

1. Architecture & tech stack (frontend frameworks, backend services, caching/CDN)
2. UI/UX & content strategy (information architecture, navigation, live-data interaction patterns)
3. Performance metrics (Mobile/Desktop performance score and LCP)
4. Developer reflection (what the comparison reveals and improvement roadmap)

3.3 Limitations

Competitor performance values can fluctuate (± 5) due to:

active ad campaigns,

A/B testing,

location-based CDN routing,

time-of-day traffic behavior.

Exact PageSpeed metrics for the PBL site were not available at the time of writing; therefore, competitor benchmarking is used to establish realistic performance expectations and improvement targets.

4. Architecture & Tech Stack Analysis (Comparative)

4.1 Sportskeeda (Hybrid Modern Delivery + Legacy CMS)

Frontend: React, Next.js (SSR), Tailwind CSS

Backend/Server: Node.js, PHP (legacy/WordPress core), Nginx

Caching/CDN: Cloudflare, Varnish

Analytics/Ads: GA4, Comscore, Prebid, Google Ad Manager

Database/Cache: MySQL (RDS), Redis

Interpretation:

Sportskeeda follows a hybrid approach: editorial publishing can remain CMS-driven (WordPress/PHP), while delivery is modernized using Next.js SSR and layered caching to improve SEO performance and scalability.

4.2 Cricbuzz (Real-Time Data Platform Architecture)

Frontend: React (migrated from AngularJS), jQuery (legacy modules)

Backend: Node.js, Java (Spring Boot), Go

Caching/CDN: Akamai, Amazon CloudFront

Infrastructure: AWS (DynamoDB, S3, ElastiCache)

Databases: MongoDB, PostgreSQL, CouchDB

Interpretation:

Cricbuzz is engineered as a real-time sports data platform. Its stack indicates an emphasis on high concurrency, efficient caching, and global edge delivery, enabling fast match-state retrieval.

4.3 ESPNcricinfo (SSR + GraphQL Content/Data Aggregation)

Frontend: React, Next.js, GraphQL (Apollo Client)

Backend: Python (Django), Node.js, Nginx

Caching/CDN: Fastly

Infrastructure: Enterprise hybrid cloud

Databases: Cassandra, MySQL, Redis

Interpretation:

GraphQL supports flexible, component-driven pages where different UI widgets request precise data, improving efficiency in large ecosystems.

4.4 Positioning of the PBL WordPress Website

The PBL site is closer to a classic CMS model: - dynamic rendering with WordPress theme templates (PHP SSR), - plugin-based performance/security controls, - REST API integration for live match features.

Key difference:

Competitors scale through distributed services + edge caching layers, while WordPress scales primarily through plugin optimization and hosting upgrades.

5. UI/UX and Content Strategy Comparison

5.1 UI Philosophy

PBL WordPress site: clean portal-style layout, custom sport templates, controlled theme consistency.

Sportskeeda: engagement-driven layout with dense content blocks, strong discovery and SEO-oriented linking, heavy ad placements.

Cricbuzz: utility-first and text-first, optimized for speed of information retrieval during live matches.

ESPNcricinfo: editorial depth + structured match pages; component-driven layouts supported by GraphQL.

5.2 Live Data UX

PBL site: REST API integration demonstrates dynamic live content, typically implemented via page-load fetch or polling with caching.

Cricbuzz-class platforms: optimized incremental updates and caching strategy to keep match pages responsive at scale.

6. Performance Metrics Evaluation (PageSpeed / Lighthouse)

6.1 Comparative Performance Table (March 2026 averages)

Website	Platform	Performance Score	LCP (Largest Contentful Paint)
Sportskeeda	Mobile	42	3.8s
Sportskeeda	Desktop	88	1.1s
Cricbuzz	Mobile	55	2.9s
Cricbuzz	Desktop	92	0.8s
SPORTLY	Mobile	60	1.9s
SPORTLY	Desktop	84	1.4s

6.2 Analysis of Differences

Mobile performance drop due to ad-tech

All three industry platforms rely heavily on programmatic advertising, which adds significant JavaScript execution and third-party requests, lowering mobile performance scores.

Cricbuzz LCP advantage

Cricbuzz's lighter, text-first UI and aggressive CDN edge caching contribute to consistently low desktop LCP.

SSR (Next.js) balancing SEO and performance

Sportskeeda and ESPNcricinfo use Next.js SSR to maintain SEO friendliness while still supporting dynamic UI components.

Expected performance behavior for WordPress project sites
WordPress performance is strongly affected by:

origin hosting limits (TTFB variability),
image formats and sizes,
theme efficiency (queries, loops, template structure),
plugin overlap and page builder output complexity.

7. Conclusion & Developer Reflection (Version 2.0 Roadmap)

7.1 Conclusion

The PBL project successfully delivered a dynamic sports website using WordPress with custom theme development, dashboard-driven content workflows, and REST API-based live cricket data integration. The competitor analysis reveals that large sports platforms operate as distributed systems where performance, scalability, and real-time update strategies are primary design constraints—often requiring SSR frameworks, multi-layer caching, and specialized backend services.

7.2 Developer Reflection (Key Learnings)

Correct architecture depends on product goals. WordPress is highly effective for CMS-driven publishing, while live scoring at scale demands distributed services.

Performance is a system property. It depends on hosting, caching strategy, assets, JavaScript execution, and third-party scripts.

Maintainability is critical. Replacing hardcoded links, using WordPress menus, and applying proper template structure are professional development practices.

7.3 Version 2.0 Improvement Plan (Actionable)

Performance Enhancements

Convert images to WebP/AVIF and enforce responsive image sizing.

Reduce unused CSS/JS and avoid heavy page-builder output where possible.

Consolidate optimization plugins to avoid conflicts and duplicate processing.

Live Match Data Optimization

Cache API responses using transients/REST cache with controlled TTL.

Implement lightweight AJAX polling for near real-time score updates.

Infrastructure Improvements

Add CDN (e.g., Cloudflare) for caching and static asset acceleration.

Upgrade origin hosting to reduce TTFB variability.

Feature Enhancements

Add user login and personalization (saved matches/teams).

Create match-centric pages (fixtures, results, series, scorecards).

Implement structured data (schema markup) for sports articles/events.

8. References

Google PageSpeed Insights / Lighthouse documentation (general performance methodology).

Wappalyzer / BuiltWith technology signature tools (stack identification methodology).

Platform stack snapshots and performance averages (March 2026) provided as project observation data:

Sportskeeda

Cricbuzz

ESPNcricinfo

9. Appendix

Appendix A — PBL Website Plugin List (Observed)

Akismet Anti-spam: Spam Protection

All-in-One WP Migration and Backup

Elementor

Hello Dolly

LiteSpeed Cache

Rank Math SEO

Site Kit by Google

Smush

Wordfence Security

WP REST Cache

WP STAGING WordPress Backup Plugin - Backup Duplicator & Migration

WP-Optimize - Clean, Compress, Cache

Appendix B — PBL Architecture Notes (Conceptual)

Custom JS + APIs + WP backend

WordPress (CMS + REST API)

Cache (transients)

Sports API

Simple flow: Frontend → Backend → Cache → API → User